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Towards Combinatorial Interpretability of Neural Computation

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Abstract

We introduce *combinatorial interpretability*, a methodology for understanding neural computation by analyzing the combinatorial structures in the sign-based categorization of a network's weights and biases. We demonstrate its power through *feature channel coding*, a theory that explains how neural networks compute Boolean expressions and potentially underlies other categories of neural network computation. According to this theory, features are computed via feature channels: unique cross-neuron encodings shared among the inputs the feature operates on. Because different feature channels share neurons, the neurons are polysemantic and the channels interfere with one another, making the computation appear inscrutable.

We show how to decipher these computations by analyzing a network's feature channel coding, offering complete mechanistic interpretations of several small neural networks that were trained with gradient descent. Crucially, this is achieved via static combinatorial analysis of the weight matrices, without examining activations or training new autoencoding networks. Feature channel coding reframes the superposition hypothesis, shifting the focus from neuron activation directionality in high-dimensional space to the combinatorial structure of codes. It also allows us for the first time to exactly quantify and explain the relationship between a network's parameter size and its computational capacity (i.e. the set of features it can compute with low error), a relationship that is implicitly at the core of many modern scaling laws.

Though our initial studies of feature channel coding are restricted to Boolean functions, we believe they provide a rich, controlled, and informative research space, and that the path we propose for combinatorial interpretation of neural computation can provide a basis for understanding both artificial and biological neural circuits.

1 Introduction

An exciting recent body of research in the interpretability of neural networks has been focused on viewing their computation as collections of independent features whose directions in high dimensional activation space can be analyzed. This has led, using novel approaches such as the use of sparse autoencoders [29], to the discovery of polysemanticity, the superposition hypothesis [12], and a body of empirical evidence of the correlation between features and directionality of neuronal activation space vectors [2, 3, 5, 7, 10, 11, 13, 29, 30]. Those techniques (surveyed in detail in the related work section) have provided ingenious ways of capturing the representation of features and some information on their interdependencies, but so far have not elucidated the full dependency graph, the underlying function being computed nor the mechanism for computing it.

In this paper, we take a combinatorial approach to interpretability, rather than one based on the geometry of the activation space. As we explain below, we focus on the computation of Boolean

expressions, a meaningful class of computation that makes it easier to take first steps into interpreting combinatorial representations. We use it to propose a theory explaining how neural networks compute (at least for Boolean formulae) by showing how features and their computation are represented as combinatorial structures encoded in the weights and biases of the trained network. In particular, this theory offers explanations of how polysemanticity and the superposition hypothesis arise through codes associated with features.

In this combinatorial view, we focus much less on the actual values involved in these learned parameters and more on the sign-based categorization (positive, negative, or zero) of those values. The two views into neural computation (the existing vector space one and the new combinatorial one) are not mutually exclusive, but we show here that the combinatorial approach, at least in some cases, provides insights into a number of important facets of interpretability that are harder to analyze with the vector space approach, including the logic of the circuits that compute the features. We also demonstrate, currently in a number of small examples, but hopefully in many more elaborate ones in the future, that the combinatorial approach can be used to extract the features and their exact computation directly from the structure of the learned parameters of the network. This is done without resorting to training sparse autoencoders, transcoders, or similar approaches [3, 10, 13, 29]. In other words, our work aims to reveal the exact low level mechanics of the computation, the *how*, not just the when and where of the relationships among features.

At the heart of our combinatorial approach is what we call the *feature channel coding hypothesis*. This hypothesis specifies both how a feature is represented (which has a natural analogue to the superposition hypothesis), as well as how computation proceeds between those features (which does not). In feature channel coding, a feature is represented in a layer of the network as a code using neurons at that layer. The individual "bits" of the code specify the neurons whose weights bring about an activation for that feature in that layer. If the codes for any pair of features at a given layer do not overlap much, the corresponding activation vectors (where precise values are taken into account) are almost-orthogonal, similar to the superposition hypothesis.

The real differentiator of feature channel coding is its theory of how computation proceeds. To compute a feature f_i, all of the features from the prior layer of the network needed for that computation are mapped to the feature code for f_i. Then, the network computes f_i individually on each neuron of the code. For example, if $f_i = x_{i_1} \wedge x_{i_2}$ (where x_{i_1} and x_{i_2} are Boolean variables) then, since $x_{i_1} \wedge x_{i_2}$ can computed as $\mathrm{ReLU}(x_{i_1} + x_{i_2} - 1)$, that computation is performed on every neuron of the code for f_i. When the right combination of features is present at the previous layer to activate f_i, each of the neurons of the code for f_i becomes active through their individual computations, and as a result, f_i itself will be active. When a feature at the prior layer is used for multiple features at the current layer, it gets mapped to each of the codes for the features it is used in. Note that the channel's code is somewhat unrelated to the function being computed by the feature - the goal of the channel's code is to bring together the inputs that need to be combined to compute this function. There may be noise on each channel due to the overlap between codes, but as long as the overlap is small, the computation still produces a result that approximates the correct code sufficiently well. Uncovering the codes for a given neural network will allow us to describe its "circuit," and to do so statically directly from the weight matrices of the network, without learning the directionality of neurons in activation space. We provide a more in depth description of feature channel coding below.

Here is a brief summary of the results of this paper. We demonstrate the effectiveness of combinatorial interpretability through contributions that can be broadly categorized into three areas:

1. *Establishing the Feature Channel Coding Hypothesis and Demonstrating its Manifestation.*

   We introduce the feature channel coding hypothesis, proposing that neural networks internally represent and compute with features using a specific, interpretable coding scheme. We provide substantial evidence supporting this hypothesis, showing its emergence in MLP networks trained via backpropagation and gradient descent. Most of this evidence is for single hidden layer networks, where we show the following problems exhibit feature channel coding:

   - Disjunctive Normal Form (DNF) Boolean formulas with only positive variables.
   - DNF formulas with both positive and negative variables.
   - Formulas consisting of just the OR of Boolean formulas.
   - A counter-intuitive example of a network learning Conjunctive Normal Form (CNF) formulas via double application of DeMorgan's Law to arrive at a DNF-like coding.
   - A one-dimensional analogue of a vision task, demonstrating the potential breadth of this coding phenomenon.

   We also demonstrate how to fully extract feature channel codes, and the computations they enable, from the weight matrices of the networks computing DNF Boolean formulas with positive variables as well as from the weight matrices of the network computing the one-dimensional vision task. These extractions provide a complete mechanistic interpretation of the network's computation directly from its weights.

2. *Quantifying Feature Channel Coding, Scaling, and the Learning Process.* We quantify the frequency with which the combinatorial structures of feature channel coding appear in the weight matrices. By demonstrating that these structures appear much more often than random, we provide further evidence for feature channel coding. We also apply this combinatorial approach to quantitatively analyze scaling laws based on feature coding. This yields, to our knowledge, the first concrete mechanistic explanation for why model accuracy can break down as problem complexity (e.g., the size of the underlying Boolean formula) increases: there is a combinatorial limit to the ability to "pack in" the necessary structures for feature channel codes, leading to representational collapse. The network also does not seem to have a "plan B" when increasing complexity makes feature channel coding no longer possible. These results suggest that tracking feature coding across scales could lead to more refined scaling laws. Furthermore, we use this quantitative approach to demonstrate how feature channel codes emerge dynamically during gradient descent training, showing a gradual structuring of weights from initial randomness. This observational capability opens avenues for quantitatively studying the impact of various training paradigms like batch normalization and dropout on code formation.

3. *Introducing a Framework for Deeper Networks.* All of the above results are for networks with a single hidden layer. To facilitate the analysis of deeper networks, we propose a novel disentangled perspective on neural network computation, building on techniques from [1]. This framework conceptualizes each layer's weight matrix (W_i) as a product of two distinct matrices: D_i, which transforms the previous layer's feature codes into a monosematic representation, and C_i, which projects each monosematic feature into its feature channel codes required by the features of the subsequent layer. Ignoring non-linearities and biases, the network's transformations can thus be viewed as a sequence $W_d \cdot \ldots \cdot W_2 \cdot W_1 = C_d D_d \cdot \ldots \cdot C_2 D_2 \cdot C_1 D_1$.